

Curriculum Vitae Yuto Lewis Terashima

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Profile

Keywords: Multiscale Simulation, Coupled FEM–MD, Quantum Computing, HPC, Tensor Networks
PhD candidate at Keio University and Research Assistant at G-QuAT (AIST), specializing in computational science. My work spans quantum algorithm development, HPC-based molecular dynamics, and computational mechanics. Through research at AIST, university labs, and corporate R&D centers, I have driven projects end to end – from theory to implementation. I am passionate about research and development that emerging computing technologies make possible.

Research Experience

April 2026 – Present

Research Fellow (DC2), Japan Society for the Promotion of Science (JSPS)

[Government Funding — JSPS]

Awarded the JSPS Research Fellowship for Young Scientists (DC2). Receives a monthly stipend and a research grant. Compensation source: Japanese government via JSPS (an Incorporated Administrative Agency under Ministry of Education, Culture, Sports, Science and Technology (MEXT)).

July 2024 – Present

Research Assistant, Global Research and Development Center for Business by Quantum-AI Technology (G-QuAT), National Institute of Advanced Industrial Science and Technology (AIST)

[Government Research Institute — AIST]

Develop quantum algorithms (eigenvalue transformation, gradient flow, tensor networks) for computational mechanics. Paid research assistant position. AIST is a National Research and Development Agency under Ministry of Economy, Trade and Industry (METI).

June 2026 – October 2026

Research Scientist, IBM Research-Tokyo Research Laboratory (TRL)

[Private Company]

Conducted research on quantum-centric supercomputing for quantum chemistry, including sample-based quantum diagonalization (SQD) using RIKEN GPU Cluster, ROQUO.

November 2025 – January 2026

Guest Researcher, Department of Mechanical Engineering, Eindhoven University of Technology (TU/e), The Netherlands

[Public University (NL)]

Joint research with Associate Professor Varvara Kouznetsova on multiscale FEM–MD methods. Hosted at a Dutch public technical university; travel and living supported by the JST Keio SPRING program.

April 2025 – March 2026

Research Fellow (Keio SPRING), Japan Science and Technology Agency (JST) / Keio University

[Government Funding — JST]

Awarded the JST SPRING Fellowship (Keio SPRING) for doctoral students. Receives a monthly stipend and a research grant. Compensation source: Japanese government via JST (a National Research and Development Agency under MEXT).

June 2023 – February 2024

Project Leader, MITOU Target Program, Quantum Computing Software Track, Information-technology Promotion Agency (IPA)

[Government Funding — IPA]

Led a 9-month government-funded software development project on quantum-annealing-based multi-drop delivery optimization (mentored by PM Dr. Ryo Tamura). Compensation source: Japanese government via IPA

(an Incorporated Administrative Agency under METI).

August 2023 – September 2023

Research Intern, Toyota Central R&D Labs., Inc.

[Private Company]

Developed tensor-train-decomposition-based solvers for homogenized thermal conduction and elasticity problems. Output: arXiv:2407.18870 (under review at *Scientific Reports*).

April 2022 – Present

Research Member, Muramatsu Lab., Keio University

[Private University]

Continuous research at the same lab through Bachelor's, Master's, and Ph.D. programs. Research on coupled FEM–MD multiscale simulation, polycarbonate mechanics, and CO₂ hydrate pellets.

Education

April 2025 – Present

Ph.D. Candidate, Graduate School of Science and Technology, Keio University

Supervisor: Associate Professor Mayu Muramatsu. Reserch Theme: Multiscale Simulation and Quantum Algorithms for Computational Mechanics.

April 2023 – March 2025

M.S. in Engineering, Graduate School of Science and Technology, Keio University

GPA: 4.00/4.00 (overall and major). Master's Thesis: "A Study on Multiscale Boundary Conditions in Coupled Homogenized FEM–MD Simulations for Atomistic-to-Continuum Modeling." Supervisor: Professor Kenji Yasuoka. Co-supervisor: Associate Professor Mayu Muramatsu.

April 2019 – March 2023

B.S. in Mechanical Engineering, Faculty of Science and Technology, Keio University

GPA: 3.34/4.00 (overall), 3.66/4.00 (major). Bachelor's Thesis: "A Study on the Structural Stability of Carbon Dioxide Hydrate Pellets Using Coupled MD–FEM Simulations." Supervisor: Associate Professor Mayu Muramatsu.

Awards

1. **Outstanding Presentation Award**, Japan Society for Computational Engineering and Science (JSCES), September 25, 2025.
Presented "Computational Homogenization Framework for Atomistic to Continuum Multiscale Boundary Conditions" at the 30th JSCES Conference (Omiya, June 2025) and received the award. Selected as one of 2 outstanding presentations from all conference presentations.
2. **JSCES Scholarship Award**, JSCES, June 4, 2025.
Received the scholarship in recognition of contributions to JSCES, supporting attendance at an IACM-organized international conference. Awarded to approximately 2 JSCES members aged 35 or under (as of April 1, 2024).
3. **Best Presentation Award**, Collaborative Research Idea Hackathon, JSCES Summer Student Conference 2024, September 16, 2024.
Co-presented "A Proposal of FEM–MD Analysis for Generation of Environmentally Compatible Micro Flying Robots" with Nozomu Magome and received the award. Selected as the best team out of 5 competing teams.

Publications – International Journals (Peer-Reviewed)

1. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. "Secant-Based Multiscale Constitutive Modeling toward Coupled Homogenized Finite Element Method and Molecular Dynamics for Polycarbonate." *Materials Today Communications*, vol. 54, 115578, June 2026.
2. Marina Kobayashi, Yuto Lewis Terashima, Takahiro Murashima, Julien Yvonnet, and Mayu Mu-

ramatsu. “Accelerating Hierarchical FEM–MD Simulations Using Clustering.” *Mechanical Engineering Journal*, vol. 13, No. 3, May 2026.

3. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Fine-Scale Structural Stability of Carbon Dioxide Hydrate Pellets under Coarse-Scale Deformation Using Multi-Scale Coupled FEM–MD Simulations.” *Materials Today Communications*, vol. 38, 108322, March 2024.

Publications – Under Review / Preprints

1. Yuto Lewis Terashima, Tadashi Kadowaki, Yohichi Suzuki, and Katsuhiro Endo. “Quantum Gradient Flow Algorithm for Symmetric Positive Definite Systems via Quantum Eigenvalue Transformation: Towards Quantum CAE.” *arXiv preprint*, arXiv:2512.09623, 2025.

Note: Research conducted as a Research Assistant at G-QuAT, AIST. Currently under review at *Physical Review Research*.

2. Yuki Sato, Yuto Lewis Terashima, and Ruho Kondo. “Efficient Computational Homogenization via Tensor Train Format.” *arXiv preprint*, arXiv:2407.18870, 2024.

Note: Research conducted during the Toyota Central R&D Labs. internship. Currently under review at *Scientific Reports*.

Publications – Japanese Journals (Peer-Reviewed)

1. Mayu Muramatsu, Yusuke Yamazaki, and Yuto Terashima. “Coupled MD–FEM Simulations for Nano-Polycrystalline Aluminium.” *Materials – The Society of Materials Science of Japan*, vol. 73, no. 8, 13602, 2024. (Review article in Japanese.)

International Conference Presentations

1. Yuto Lewis Terashima, Tadashi Kadowaki, Yohichi Suzuki, and Katsuhiro Endo. “Quantum Gradient Flow Algorithm for Reducing Circuit Complexity in Matrix Inverse Calculations of Finite Element Analysis.” 17th World Congress on Computational Mechanics / 10 th European Congress on Computational Methods in Applied Sciences and Engineering, Munich, Germany, July 20, 2026.
2. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Dynamic Multiscale Modeling of Polycarbonate Using Coupled Homogenized FEM–MD Simulations.” 18th U.S. National Congress on Computational Mechanics, Chicago, USA, July 22, 2025.
3. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Molecular Dynamics Calculations for Localized Shear Properties of Polycarbonate Using Computational Homogenization Framework.” 20th International Conference on Strength of Materials (ICSMA 20), Kyoto, Japan, June 2, 2025.
4. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “A Coupled Simulation of Particle Dynamics and Finite Element Analysis: Atomistic to Continuum Modeling.” First Imperial–Keio Joint Workshop, London, UK, March 2025.
5. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “A Study of Continuum-Scale Stress Calculation on Finite Element Method Induced by Molecular-Scale Structural Transition.” 16th World Congress on Computational Mechanics / 4th Pan American Congress on Computational Mechanics, Vancouver, Canada, July 23, 2024.
6. Yuto Lewis Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “An Investigation of Structural Stability of Carbon Dioxide Hydrate Pellets for Carbon Neutral Applications via Coupled MD–FEM.” 17th U.S. National Congress on Computational Mechanics, Albuquerque, USA, July 27, 2023.

Domestic Conference Presentations

1. Yuto Terashima, Varvara Kouznetsova, and Mayu Muramatsu. “Secant and Tangent-based Formulation of Atomistic to Continuum Homogenization.” 31st JSCES Conference, Nagoya, June 5, 2026.
2. Yuto Terashima, Tadashi Kadowaki, Yohichi Suzuki, and Katsuhiro Endo. “Matrix Inversion via Relaxation Process under Quantum Computational Constraints and Applications to Computational Mechanics.” 31st JSCES Conference, Nagoya, June 3, 2026.
3. Yuto Terashima, Tadashi Kadowaki, Yohichi Suzuki, and Katsuhiro Endo. “Linear System Solving via the Quantum Gradient Flow Algorithm and Its Challenges in Condition-Number-Dependent Problems.” 203rd HPC / 17th QS Joint Workshop, Hokkaido University, Sapporo, March 17, 2026.
4. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Multiscale Constitutive Modeling via Coupled Homogenized FEM and MD Enabled by Large-Scale Parallel Computing.” M&M/CMD Symposium for Young Researchers 2025, Nagoya, August 28, 2025.
5. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Computational Homogenization Framework for Atomistic to Continuum Multiscale Boundary Conditions.” 30th JSCES Conference, Omiya, June 5, 2025. (Outstanding Presentation Award.)
6. Yuto Terashima, Tadashi Kadowaki, Yohichi Suzuki, Mayu Muramatsu, and Katsuhiro Endo. “Efficient Nonlinear Finite Element Methods Using Tensor Networks and Their Verification.” 198th HPC / 14th QS Joint Workshop, Hokkaido University, Sapporo, March 17, 2025.
7. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Homogenized Dynamic FEM–MD Simulation for Deformation Analysis of Polycarbonate.” JSME 37th Computational Mechanics Division Conference, Sendai, October 18, 2024.
8. Yuto Terashima, Paul Edward Brumby, Varvara Kouznetsova, and Mayu Muramatsu. “A Method for Reflecting Continuum-Scale Element Deformation from Molecular-Scale Structural Transition.” 29th JSCES Conference, Kobe, June 12, 2024.
9. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, Varvara Kouznetsova, and Mayu Muramatsu. “Macroscopic Deformation Analysis of Hydrates by FEM–MD Multiscale Simulation.” 37th Molecular Simulation Symposium, Fukui, December 5, 2023.
10. Yuto Terashima, Paul Edward Brumby, Varvara Kouznetsova, and Mayu Muramatsu. “Macroscopic Mechanical Properties of Hydrates via Coupled FEM–MD Simulation.” JSME 36th Computational Mechanics Division Conference, Toyohashi, October 26, 2023.
11. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, and Mayu Muramatsu. “Microscopic Structural Stability of Hydrate under Macroscopic Deformation.” 28th JSCES Conference, Tsukuba, June 2, 2023.
12. Yuto Terashima, Paul Edward Brumby, Takahiro Murashima, and Mayu Muramatsu. “Homogenized Structural Analysis of Hydrate via MD–FEM.” 36th Molecular Simulation Symposium, Tokyo Institute of Technology, December 5, 2022.

Invited Talks

1. Yuto Terashima. “Multiscale Analysis from Microscopic Physics, and Quantum Computing.” Industry and Career Transformation through Molecular Simulation, Tech Hub Yokohama, Yokohama, October 31, 2025.
2. Yuto Terashima. “Multiscale Finite Element Analysis Using Supercomputers and Quantum Computers.” 14th Multi-Method / Novel Numerical Simulation Method Conference, Online, September 30, 2025.
3. Yuto Terashima. “Ph.D. Student Invited Talk 4: Quantum–Atomistic–Continuum Multiscale Simulation.” JSCES Summer Student Conference 2025, Tsukuba, September 20, 2025.

Patents

1. JP Patent Application 2025-015426 / Patent No. 7739661. “Delivery Planning Device, Delivery Planning System, Delivery Planning Method, and Program.” Applicant: Keio University. Inventors: Rio Honda, Satoki Ishiai, Yuto Terashima, and Shiori Aoki.

Other Achievements

1. Yuto Terashima, Satoki Ishiai, Rio Honda, and Shiori Aoki. AY2023 IPA MITOU Target Program (Quantum Computing Software Track). Project: “Development of a Novel Delivery Optimization Method via Annealing Machines.” Selected under PM Dr. Ryo Tamura. Led the project for 9 months as team leader, planning the development concept and driving software development on a quantum annealing machine. June 2023 – February 2024.

Committee Service

April 2025 – September 2025

Student Organizing Committee Member, JSCES Summer Student Conference 2025

Participated in planning meetings and on-site preparations before and during the conference.

Extracurricular Activities

August 2025

Participant, Global Citizenship Program (New York), Keio University

Conducted a survey project on the New York innovation ecosystem, including visits to IBM Research and Rensselaer Polytechnic Institute (RPI).

July 2025

Volunteer (Student Chapter Member), USACM at the 18th U.S. National Congress on Computational Mechanics

Helped organize the poster session and registration in Chicago.

November 2024

Organizing Committee, Nosé 40 Workshop

Helped organize a workshop on phase-space sampling using molecular simulation, celebrating the 40th anniversary of Nosé dynamics.

August 2024

Participant, Qiskit Hackathon Taiwan 2024

Developed a fast variational quantum eigensolver to calculate chemical potential.

July 2023

Volunteer (Student Chapter Member), USACM at the 17th U.S. National Congress on Computational Mechanics

Helped organize the poster session and registration in Albuquerque.